MILITARY SPECIFICATION SHEET

FLANGES, WAVEGUIDE (PRESSURIZABLE CONTACT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the flanges described herein shall consist of this document and the latest issue of Specification MIL-F-3922.

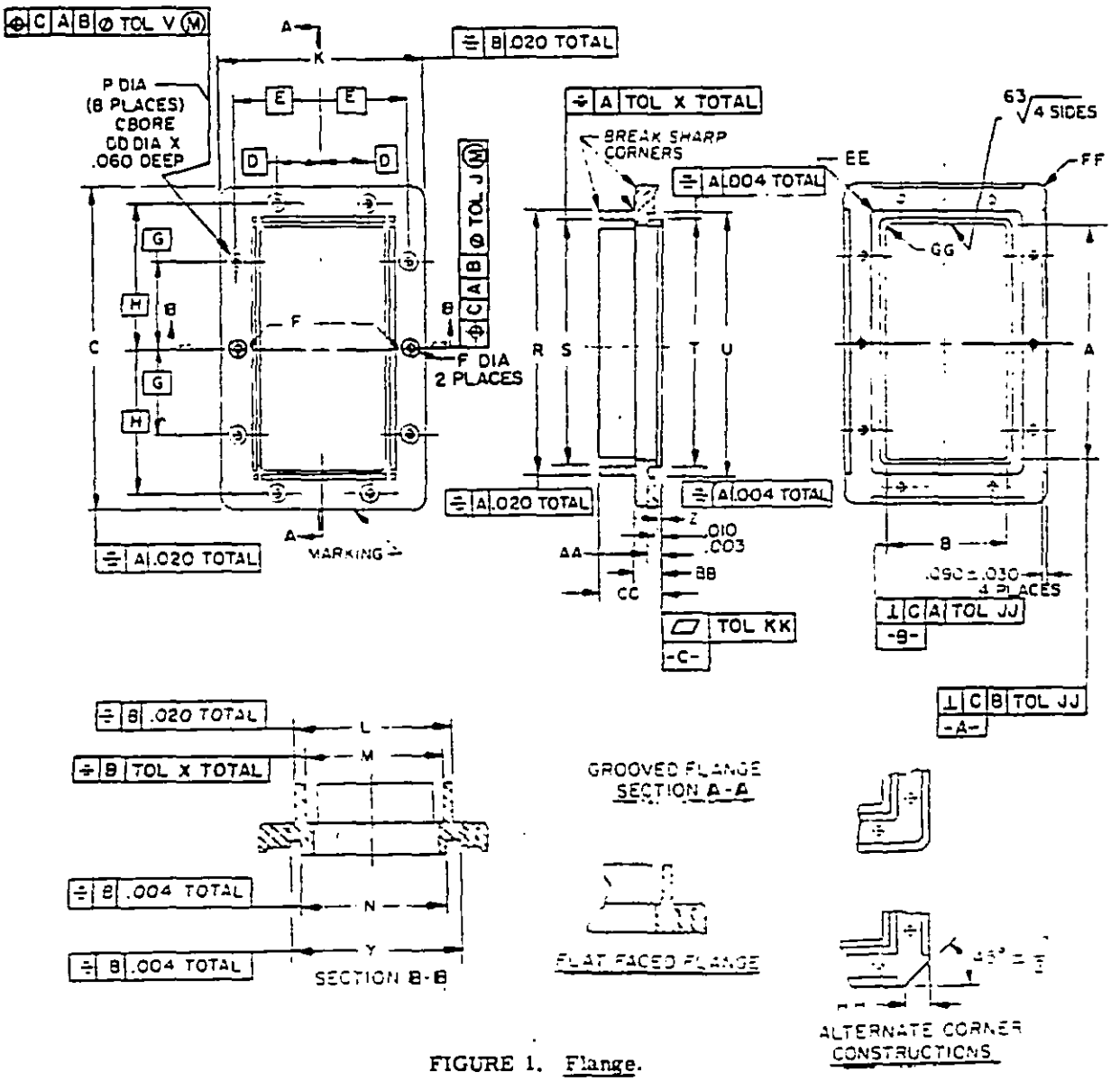


FIGURE 1. Flange.

FSC 5985

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Mating surfaces, including seal groove, shall not exceed 63 microinches in accordance with ANSI - B46.1 - 1962, except that flaws shall be included in the roughness height measurement.
4. Flat face flanges require a metal-rubber seal at the junction. These seals are covered by MIL-G-24211.
5. For lettered dimensions, see table I.

INCHES	MM
.003	.08
.004	.10
.010	.25
.020	.51
.030	.76
.040	1.02
.060	1.52
.090	2.29

FIGURE 1. Flange - Continued.

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TABLE I. Dash numbers and dimensions.

Part No. M3922/52- 1/						Dimensions 2/						
	Used with			Used with		A	B	C ±.02 (.51)	D BSC.	E BSC.	F Dia.	G BSC.
Dash No. 3/	Waveguide M85/1-	Mating flange	Dash No. 4/	Waveguide M85/1-	Mating flange							
001	015	5/	002	018, 019,	5/	6.500 ±.005	3.250 ±.005	8.69	1.249	2.311	.330 ±.005	2.374
023	017		024	168		(165.10) (.13)	(82.55) (.13)	(220.73)	(31.72)	(58.70)	(8.38) (.13)	(60.30)
003	021		004	025, 026,		5.100 ±.005	2.550 ±.005	7.16	1.031	1.925	.266 ±.005	1.987
025	023		026	169		(129.54) (.13)	(64.77) (.13)	(181.86)	(26.19)	(48.90)	(6.76) (.13)	(25.07)
005	027		006	029, 030,		4.300 ±.005	2.150 ±.005	6.34	.938	1.720	.266 ±.005	1.787
027	031		028	170		(109.22) (.13)	(54.61) (.13)	(161.04)	(23.82)	(43.69)	(6.76) (.13)	(45.39)
007	033		008	035, 036,		3.400 ±.005	1.700 ±.005	5.44	.671	1.500	.266 ±.005	1.344
029	037		030	171		(86.36) (.13)	(43.18) (.13)	(138.18)	(17.04)	(38.10)	(6.76) (.13)	(34.14)
009	039		010	041, 042,		2.840 ±.005	1.340 ±.005	4.50	.578	1.164	.257 ±.003	1.281
031	043		032	172		(72.14) (.13)	(34.04) (.13)	(114.30)	(14.68)	(29.57)	(6.53) (.08)	(32.54)
011	045		012	047, 048,		2.290 ±.005	1.145 ±.005	3.88	.500	1.050	.257 ±.002	1.070
033	049		034	173		(58.17) (.13)	(29.08) (.13)	(98.55)	(12.70)	(26.67)	(6.53) (.05)	(27.18)
013	051		014	053, 054,		1.872 ±.005	.872 ±.005	3.50	.438	.914		.563
035	055		036	174		(47.55) (.13)	(22.15) (.13)	(88.90)	(11.12)	(23.22)		(14.30)
015	057		016	059, 060,		1.590 ±.004	.795 ±.004	3.19	.375	.875		.500
037	061		038	175		(40.39) (.10)	(20.19) (.10)	(81.03)	(9.52)	(22.22)		(12.70)
017	063		018	065, 066,		1.372 ±.004	.622 ±.004	2.69	.313	.719		.438
039	067		040	176		(34.85) (.10)	(15.80) (.10)	(68.33)	(7.95)	(18.26)		(11.12)
019	069		020	071, 072,		1.122 ±.004	.497 ±.004	2.50	.313	.640		.375
041	073		042	177		(28.50) (.10)	(12.62) (.10)	(63.50)	(7.95)	(16.26)		(9.52)
021	075		022	077, 078,		.900 ±.003	.400 ±.003	2.09	.313	.578		.312
043	079	5/	044	178	5/	(22.86) (.08)	(10.16) (.08)	(53.09)	(7.95)	(14.68)		(7.92)

See footnotes at end of table.

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TABLE I. Dash numbers and dimensions - Continued.

Dimensions 2/ (Continued)												
Dash No. 3/	Dash No. 4/	H BSC.	J	K ±.02 (.51)	L ±.02 (.51)	M	N ±.004 (.10)	P Dia.	R ±.02 (.51)	S	T ±.004 (.10)	U ±.004 (.10)
001 023	002 024	3.937 (100.00)	.012 (.30)	5.44 (138.18)	3.81 (96.77)	3.420 -.005 (86.87) (.13) +.000	3.630 (92.20)	.330 ±.005 (8.38) (.13)	7.06 (179.32)	6.670 -.005 (169.42) (.13) +.000	6.880 (174.75)	7.452 (189.28)
003 025	004 026	3.200 (81.28)	.011 (.28)	4.62 (117.35)	3.10 (78.74)	2.720 -.005 (69.09) (.13) +.000	2.920 (74.17)	.266 ±.005 (6.76) (.13)	5.67 (144.02)	5.270 -.005 (133.86) (.13) +.000	5.470 (138.94)	6.042 (153.47)
005 027	006 028	2.795 (71.00)	.011 (.28)	4.19 (106.43)	2.69 (68.33)	2.320 -.005 (58.93) (.13) +.000	2.510 (63.75)	.266 ±.005 (6.76) (.13)	4.88 (123.95)	4.470 -.005 (113.54) (.13) +.000	4.660 (118.36)	5.232 (132.89)
007 029	008 030	2.344 (59.54)	.011 (.28)	3.75 (95.25)	2.25 (57.15)	1.870 -.005 (47.50) (.13) +.000	2.062 (53.37)	.266 ±.005 (6.76) (.13)	4.00 (101.60)	3.570 -.005 (90.68) (.13) +.000	3.782 (96.06)	4.354 (110.59)
009 031	010 032	1.914 (48.62)	.004 (.10)	3.00 (76.20)	1.75 (44.45)	1.510 -.005 (38.35) (.13) +.000	1.624 (41.25)	.257 +.005 (6.53) (.13) -.000	3.25 (82.55)	3.010 -.005 (76.45) (.13) +.000	3.124 (79.35)	3.500 (88.90)
011 033	012 034	1.620 (41.15)	.005 (.13)	2.75 (69.85)	1.50 (38.10)	1.283 -.005 (32.59) (.13) +.000	1.395 (35.43)	.257 +.005 (6.53) (.13) -.000	2.63 (66.80)	2.428 -.005 (61.67) (.13) +.000	2.540 (64.52)	2.916 (74.07)
013 035	014 036	1.414 (35.92)	---	2.50 (63.50)	1.25 (31.75)	1.010 -.005 (25.65) (.13) +.000	1.122 (28.50)	.257 +.005 (6.53) (.13) -.000	2.25 (57.15)	2.010 -.005 (54.05) (.13) +.000	2.122 (53.90)	2.498 (63.45)
015 037	016 038	1.273 (32.33)	---	2.44 (61.90)	1.13 (28.70)	.931 -.004 (23.64) (.10) +.000	1.045 (26.54)	.257 +.005 (6.53) (.13) -.000	1.94 (49.28)	1.726 -.004 (43.84) (.10) +.000	1.840 (46.74)	2.216 (56.29)
017 039	018 040	1.094 (27.79)	---	1.94 (49.28)	1.00 (25.40)	.758 -.004 (19.25) (.10) +.000	.872 (22.15)	.196 +.004 (4.98) (.10) -.000	1.75 (44.45)	1.508 -.004 (38.30) (.10) +.000	1.622 (41.20)	1.904 (48.36)
019 041	020 042	.953 (24.21)	---	1.75 (44.45)	.88 (22.35)	.633 -.004 (16.08) (.10) +.000	.747 (18.97)	.169 +.002 (4.29) (.05) -.000	1.50 (38.10)	1.258 -.004 (31.95) (.10) +.000	1.372 (34.85)	1.654 (42.01)
021 043	022 044	.828 (21.03)	---	1.59 (40.39)	.75 (19.05)	.506 -.003 (12.86) (.08) +.000	.624 (15.85)	.169 +.002 (4.29) (.05) -.000	1.38 (35.05)	1.006 -.003 (25.55) (.08) +.000	1.124 (28.55)	1.406 (35.71)

See footnotes at end of table.

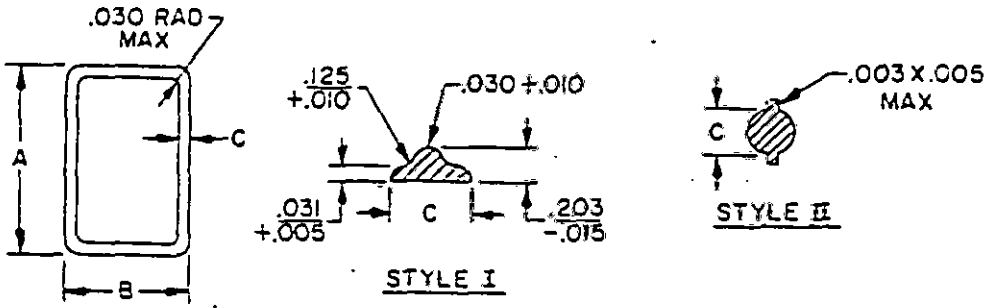
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TABLE I. Dash numbers and dimensions - Continued.

Dimensions 2/ (Continued)															
Dash No. 3/	Dash No. 4/	V	X	Y ±.0040 (.10)	Z Max	AA ±.005 (.13)	BB ±.02 (.51)	CC ±.02 (.51)	DD ±.005 (.13)	EE ±.003 (.08)	FF ±.02 (.51)	GG Max	HH ±.02 (.51)	JJ	KK
001	002	.012	.003	4.202	.005	.062	.50	.88	.594	.143	.25	.05	.25	.005	.005
023	024	(.30)	(.08)	(106.73)	(.13)	(1.57)	(12.70)	(22.35)	(15.09)	(3.63)	(6.35)	(1.27)	(6.35)	(.13)	(.13)
003	004	.011	.003	3.492	.005	.062	.50	.88	.531	.143	.25	.05	.25	.005	.005
025	026	(.28)	(.08)	(88.70)	(.13)	(1.57)	(12.70)	(22.35)	(13.49)	(3.63)	(6.35)	(1.27)	(6.35)	(.13)	(.13)
005	006	.011	.003	3.082	.005	.062	.50	.88	.531	.143	.25	.05	.25	.005	.005
027	028	(.28)	(.08)	(78.28)	(.13)	(1.57)	(12.70)	(22.35)	(13.49)	(3.63)	(6.35)	(1.27)	(6.35)	(.13)	(.13)
007	008	.011	.003	2.634	.005	.062	.50	.88	.531	.143	.25	.05	.25	.005	.005
029	030	(.28)	(.08)	(66.90)	(.13)	(1.57)	(12.70)	(22.35)	(13.49)	(3.63)	(6.35)	(1.27)	(6.35)	(.13)	(.13)
009	010	.007	.003	2.000	.002	.053	.38	.69	.531	.094	.31	.05	.31	.005	.005
031	032	(.18)	(.08)	(50.80)	(.05)	(1.35)	(9.65)	(17.53)	(13.49)	(2.39)	(7.87)	(1.27)	(7.87)	(.13)	(.13)
011	012	.007	.003	1.771	.002	.053	.38	.69	.531	.094	.31	.05	.31	.005	.005
033	034	(.18)	(.08)	(44.98)	(.05)	(1.35)	(9.65)	(17.53)	(13.49)	(2.39)	(7.87)	(1.27)	(7.87)	(.13)	(.13)
013	014	.007	.003	1.498	.002	.053	.38	.69	.531	.094	.31	.03	.31	.005	.005
035	036	(.18)	(.08)	(38.05)	(.05)	(1.35)	(9.65)	(17.53)	(13.49)	(2.39)	(7.87)	(.76)	(7.87)	(.13)	(.13)
015	016	.007	.002	1.421	.002	.053	.38	.69	.531	.094	.31	.03	.31	.004	.004
037	038	(.18)	(.05)	(36.09)	(.05)	(1.35)	(9.65)	(17.53)	(13.49)	(2.39)	(7.87)	(.76)	(7.87)	(.10)	(.10)
017	018	.006	.002	1.154	.002	.040	.25	.50	.438	.070	.25	.03	.25	.004	.004
039	040	(.15)	(.05)	(29.31)	(.05)	(1.02)	(6.35)	(12.70)	(11.31)	(1.78)	(6.35)	(.76)	(6.35)	(.10)	(.10)
019	020	.005	.002	1.029	.002	.040	.25	.50	.406	.070	.25	.03	.25	.004	.004
041	042	(.13)	(.05)	(26.14)	(.05)	(1.02)	(6.35)	(12.70)	(10.31)	(1.78)	(6.35)	(.76)	(6.35)	(.10)	(.10)
021	022	.005	.002	.906	.002	.040	.25	.50	.406	.070	.25	.03	.25	.003	.003
043	044	(.13)	(.05)	(23.01)	(.05)	(1.02)	(6.35)	(12.70)	(10.31)	(1.78)	(6.35)	(.76)	(6.35)	(.08)	(.08)

1/ Part numbers 001 through 022 are for flat flanges; part numbers 023 through 044 are for grooved flanges.
2/ Metric equivalents are in parentheses.
3/ Copper alloy.
4/ Aluminum alloy.
5/ Each flange mates with an identical part number.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. One seal is to be supplied with each grooved flange.

INCHES	MM	INCHES	MM
.003	.08	.125	3.18
.005	.13	.164	4.17
.010	.25	.190	4.83
.015	.38	.203	5.16
.030	.76	.250	6.35
.031	.79	.3125	7.94
.040	1.02	.625	15.88
.047	1.19	.75	19.05
.062	1.57	1.0	25.40
.078	1.98	1.5	38.10

TABLE II. Gasket dimensions.

Style	Flange used with M3922/52-	Dimensions 1/ A	B ±.010 (.25)	C	Style	Flange used with M3922/52-	Dimensions 1/ A	B ±.010 (.25)	C
I	023 - 024	7.454 ±.015 (189.33) (.38)	4.204 (106.78)	.288 -.010 -.000 (7.32) (.25)	II	035 - 036	2.449 ±.010 (62.20) (.25)	1.449 (36.80)	.139 ±.004 (3.53) (.10)
I	025 - 026	6.044 ±.015 (153.52) (.38)	3.494 (88.75)	.288 -.010 -.000 (7.32) (.25)	II	037 - 038	2.167 ±.010 (55.04) (.25)	1.372 (34.85)	.139 ±.004 (3.53) (.10)
I	027 - 028	5.234 ±.015 (132.94) (.38)	3.084 (78.33)	.288 -.010 -.000 (7.32) (.25)	II	039 - 040	1.866 ±.010 (47.40) (.25)	1.116 (28.35)	.103 ±.003 (2.62) (.08)
I	029 - 030	4.336 ±.015 (110.13) (.38)	2.636 (66.95)	.288 -.010 -.000 (7.32) (.25)	II	041 - 042	1.616 ±.010 (41.05) (.25)	.991 (25.17)	.103 ±.003 (2.62) (.08)
II	031 - 032	3.451 ±.015 (87.66) (.38)	1.951 (49.56)	.139 ±.004 (3.53) (.10)	II	043 - 044	1.368 ±.010 (34.75) (.25)	.866 (22.05)	.103 ±.003 (2.62) (.08)
II	033 - 034	2.867 ±.010 (72.82) (.25)	1.722 (43.74)	.139 ±.004 (3.53) (.10)					

1/ Metric equivalents are in parentheses.

FIGURE 2. Grooved flange seal.

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REQUIREMENTS:

Dimensions and configuration: Flange - See figure 1 and table I.
Gasket - See figure 2.

Material: Flange - Metallic base alloy as specified in table I.

Items supplied with flanges: See table III.

Marking: See figure 1.
Part number: M3922/52- (and dash number from table I.

AN nomenclature and engineering data: See table IV.

TABLE III. Items supplied with flange.

Part number 1/ M3922/52-	Hexagon head cap screw	Hexagon nut	Spring lockwasher	Quantity (each)
001, 002, 023, 024	(5/16) .3125-18 UNC-2A, 1.5 inch long	(5/16) .3125-18 UNC-2B	.3125 x .078	5
003, 004, 005, 006, 007, 008, 025, 026, 027, 028, 029, 030	(1/4) .250-20 UNC-2A, 1.5 inch long	(1/4) .250-20 UNC-2B	.250 x .062	5
009, 010, 011, 012, 031, 032, 033, 034	(1/4) .250-20 UNC-2A, 1.0 inch long	(1/4) .250-20 UNC-2B	.250 x .062	5
013, 014, 015, 016, 035, 036, 037, 038	(1/4) .250-20 UNC-2A, 1.0 inch long	(1/4) .250-20 UNC-2B	.250 x .062	4
017, 018, 039, 040	(No. 10) .190-32 UNF-2A, .75 inch long	(No. 10) .190-32 UNF-2B	.190 x .047	4
019, 020, 021, 022, 041, 042, 043, 044	(No. 8) .164-32 UNC-2A, .750 inch long	(No. 8) .164-32 UNC-2B	.164 x .040	4

1/ One gasket (see figure 2) shall be supplied with each part number 023 through 044.

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TABLE IV. Cross-reference of part numbers/AN nomenclature and engineering data.

Part No. M3922/ 52-	AN nomenclature	Frequency range
		GHz
001	UG-1714/U	1.12 - 1.70
002	UG-1720/U	1.12 - 1.70
003	UG-1715/U	1.45 - 2.20
004	UG-1717/U	1.45 - 2.20
005	UG-1716/U	1.70 - 2.60
006	UG-1711/U	1.70 - 2.60
007	UG-1712/U	2.20 - 3.30
008	UG-1713/U	2.20 - 3.30
009	UG-1724/U	2.60 - 3.95
010	UG-1725/U	2.60 - 3.95
011	UG-1726/U	3.30 - 4.90
012	UG-1727/U	3.30 - 4.90
013	UG-1728/U	3.95 - 5.85
014	UG-1729/U	3.95 - 5.85
015	UG-1730/U	4.90 - 7.05
016	UG-1731/U	4.90 - 7.05
017	UG-1732/U	5.85 - 8.20
018	UG-1733/U	5.85 - 8.20
019	UG-1734/U	7.05 -10.0
020	UG-1735/U	7.05 -10.0
021	UG-1736/U	8.20 -12.40
022	UG-1737/U	8.20 -12.40
023	UG-1362/U	1.12 - 1.70
024	UG-1343/U	1.12 - 1.70
025	UG-1718/U	1.45 - 2.20
026	UG-1719/U	1.45 - 2.20
027	UG-1344/U	1.70 - 2.60
028	UG-1345/U	1.70 - 2.60
029	UG-1346/U	2.20 - 3.30
030	UG-1347/U	2.20 - 3.30
031	UG-1348/U	2.60 - 3.95
032	UG-1349/U	2.60 - 3.95
033	UG-1350/U	3.30 - 4.90
034	UG-1351/U	3.30 - 4.90
035	UG-1352/U	3.95 - 5.85
036	UG-1353/U	3.95 - 5.85
037	UG-1354/U	4.90 - 7.05
038	UG-1355/U	4.90 - 7.05
039	UG-1356/U	5.85 - 8.20
040	UG-1357/U	5.85 - 8.20
041	UG-1358/U	7.05 -10.0
042	UG-1359/U	7.05 -10.0
043	UG-1360/U	8.20 -12.40
044	UG-1361/U	8.20 -12.40

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Navy - EC
Air Force - 85

Review activities:
Army - MI, MU, SG, EL
Navy - None
Air Force - 11, 85, 80
DSA - ES

User activities:
Army - AV
Navy - AS, OS, MC, CG, SH
Air Force - 14, 19

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Preparing activity:
Navy - EC

Agent:
DSA - ES

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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